

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **Distributed System**

Code No: **CACS352** Semester: **VI**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. "Distributed system acts as a single coherent system to its end user." Justify the statement with its features and challenges.
3. Differentiate between client server and peer to peer architecture in distributed system.
4. Describe transient synchronous communication with suitable diagram.
5. Explain wrapper and interceptor in detail.
6. Describe the working mechanism of distributed commit.
7. Discuss the different attacks that can target distributed systems and explain how they can be mitigated.
8. Write short note (Any two)a) DNS Query Typesb) Checkpointingc) Digital Signature

Group C

Attempt any TWO questions[2x10=20]

9. What do you mean by logical clock? Explain Lamport's Logical clock.
10. How can you claim that replication is one of the scaling techniques in distributed system? Explain consistency protocols with suitable diagram.
11. What is virtualization? Explain different types of virtualizations with example. Explain the function of middleware in distributed system with suitable diagram.